

**Strategies for knowledge acquisition in bionanotechnology:
why are interdisciplinary practices less widespread than expected?**

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Abstract

Discourses on convergent technologies claim that fields such as bionanotechnology are interdisciplinary and therefore require specific organisational forms (e.g. laboratories with researchers from many different disciplinary backgrounds). However, empirical investigations have challenged the intrinsic interdisciplinarity of these emergent fields (e.g. Schummer, 2004), whilst some analysts have critiqued the discourses on as prescriptive (Weingart, 2000). In order to investigate actual laboratory practices in bionanoscience, we explored the dynamics of knowledge integration and the knowledge acquisition strategies for ten research projects in two research specialties (biomolecular motors and lab-on-a-chip). First, we found that knowledge integration was very asymmetrical: a project used materials and techniques from various disciplines at standard level of know-how, but focused its research effort on the unique expertise of the home laboratory. Second, projects displayed various strategies to acquire knowledge: only two of the ten projects examined relied mainly on ‘fully’ interdisciplinary practices (deep collaborations across laboratories based in different disciplines or within laboratories containing many backgrounds), while most projects combined different strategies, including service collaboration, limited recruitment and in-house learning. We argue that these observations may be explained by a trade-off between the benefits of cognitive diversity against the costs of team cohesion and learning.

1. Introduction

One of the central tenets of the current discourse on innovation is that the most important scientific and technological breakthroughs are the result of interdisciplinary endeavours and, therefore, innovation policies should facilitate and foster interdisciplinary research. However, a quick look into interdisciplinarity ‘in practice’ shows that it is extremely diverse: interdisciplinary research is carried out through of a variety of practices. What are these practices and how do they differ? In this article we address this question by investigating the strategies that laboratories use in order to acquire bodies of knowledge beyond their domestic expertise for two research specialties in bionanoscience.

The idea of interdisciplinarity as a source of creativity and innovation has become particularly relevant in those areas of science and technology (S&T) that have emerged as a result of technological convergence, such as nanotechnology (Malsch, 1997) or NBIC technologies (Nano-, Bio, Info-, and Cognitive-based) (Roco and Bainbridge, 2002). The growing importance of interdisciplinarity is also one of the central claims of a number of policy studies arguing that the science and technology system is undergoing major structural changes (e.g. Gibbons et al. 1994; Leydesdorff and Etzkowitz, 1998; see Hessels and van Lente, 2007 for a recent review). In parallel with these discourses, since the early 1990s there has been a boom in the (self-reported) adoption of interdisciplinarity by both scientists and policy-makers (Braun and Schubert, 2003). In both discourses and policies, interdisciplinarity is presented as the result of particular strategies of knowledge acquisition: (i) research collaboration among two or more laboratories affiliated with different disciplines, (ii) or recruitment in one laboratory of researchers from many different backgrounds (Gibbons et al. 2004, pp. 4-6, 147-150). Dr Jerry Salomon of the Caltech Beckman Institute expressed the latter view vividly:

You know Beckman’s idea was to put chemists and biologists together and throw in some computing stuff and see what came out. I think [interdisciplinarity] it’s really no more, no less than that.

Salomon quoted in Scerri (2000, p. 202).

However, social studies of science literature suggests that the practices that lead to interdisciplinary research may be more diverse. It is widely acknowledged that interdisciplinary research ‘does not necessarily imply collaboration between researchers from different disciplines’ (Bordons et al. 2004, p. 440), and even when it does, the term collaboration encompasses a variety of practices (Laudel, 2001, 2002).

Moreover, the strong normative component of much of the discourse on interdisciplinarity suggests that the discourse on interdisciplinarity ‘is, in effect, a discourse on innovation in knowledge production’ (Weingart, 2000, p. 30) that ‘provides a means for steering and coordinating strategic investment in research across a range of partners’ (Lowe and Phillipson, 2006, p. 167). In other words, the rhetoric on the benefits of interdisciplinarity is a convenient tool to foster reform in scientific institutions (e.g. tenure track, see Fuller, 2003) –whereas generally the argument is presented in the reverse sense: that organisational reform is necessary to support interdisciplinarity. This reversal is what, in his critique of Mode 2, Godin refers to as performative discourse: ‘suggesting a new organization of knowledge and participating in its realization’ (Godin, 1998, p. 465).

These arguments about the difference between discourse and practice in interdisciplinarity received some support from the results of a bibliometric study on nanotechnology conducted by Schummer (2004): although nanotechnology has been presented as an intrinsically interdisciplinary endeavour, most nanotechnology specific journals publish articles by authors from one single disciplinary affiliation, e.g. *Nanotechnology* mainly publishes articles written by physicists. However, bibliometric studies can only reveal one strategy for interdisciplinary knowledge acquisition: collaboration between departments from different disciplines. Could it be that laboratories rely on other strategies? In a previous paper we assessed degree of interdisciplinarity in one research specialty in bionanotechnology (biomolecular motors) analysing bibliometric indicators and laboratory practices (Rafols and Meyer, 2007). In some cases the contents of research was found to be interdisciplinary even when the

projects were affiliated to or carried out by researchers from a single discipline. But then, how did these projects managed to garner different bodies of knowledge?

In order to understand the variety of interdisciplinary practices, in this article we explore the knowledge acquisition strategies used by academic laboratories in ten case studies for two research specialties in bionanotechnology, biomolecular motors (BM) and lab-on-a-chip (LOC). What are the practices for knowledge integration and acquisition? In what way are these practices interdisciplinary? How and why do they differ between research specialties?

The findings presented suggest that the practices more associated with interdisciplinarity in science policy and management, such as deep collaboration between laboratories or the transient recruitment of a whole mix of researchers from diverse backgrounds, are much less common than might be expected from the policy discourse on nanotechnology. Instead, laboratories seem to rely on their core expertise and the combination of various strategies, such as limited recruitment, service collaboration or in-house development. The choice of strategy depends on the laboratories' goals (more educational or more research-focused), institutional conditions (such as funding or student availability), and capabilities to absorb knowledge from other areas. We argue that this choice can be understood in terms of a trade-off between the benefits of cognitive diversity and the costs incurred of maintaining team's cohesion or achieving the desired learning (Llerena and Meyer-Krahmer, 2004; Hollingsworth, 2006).

2. Conceptual and methodological framework

The definition of disciplinary and interdisciplinary research is problematic and controversial. This study follows the sociology of science literature on the dynamics of research and disciplines (Weingart, 2000; Becher and Trowler, 2001), which sees disciplines as social constructs with tightly associated cognitive dimensions ('tribes with territories' as Becher puts it), but highlights that the actual arena of research is the research specialty, i.e. the body of knowledge constituted around specific objects,

problems, methodologies and/or concepts that advances through the contributions of an identifiable community (the invisible college in Crane's (1972) terms). In accordance with extensive practice among scholars (Klein, 2000, p. 17; Porter et al., 2006, p. 3), here we define interdisciplinarity as a mode of research that integrates different bodies of knowledge, i.e. as a cognitive characteristic of the research process¹.

Since we are interested in the practices² that lead to interdisciplinarity, we need to define the units of analysis for both the (potentially integrative) research outcome and the social spheres mobilised. For the research outcome, we take as unit of analysis the 'research project' which is defined as a set of persistent practices aimed at solving a scientific or technological problem. The research project is then identified (or 'operationalised') via scientific contributions in the form of a sequence of publications that show coherence in terms of the topics addressed and the main researchers involved, over a particular time span (2-5 years).³ As unit of analysis for social space, we take the 'individual laboratory', which is regarded as the platform where research is conducted based on a 'domestic' knowledge embedded in its instruments, researchers and local interactive networks. In line with this perspective, in this study we use the term collaboration to refer only to investigations that involve researchers based in different laboratories.⁴

The goal of this paper is to provide an understanding of how in a given project a laboratory acquires knowledge from different sources. Or, in other words, to investigate the mobilisation of social resources in order to gain the cognitive diversity needed for a research objective. We want to uncover the knowledge acquisition strategies (social processes) through which knowledge integration (a cognitive process) is achieved. Based

¹ The term interdisciplinarity is used here to represent all types of research crossing knowledge boundaries, including multi-, trans-, and cross-disciplinarity.

² Throughout the text we use the term 'laboratory practice' to denote activities developed by researchers in a laboratory to gather and produce knowledge. Our usage is more organisational and less micro- than in classical laboratory studies, e.g. in Knorr-Cetina's *Epistemic cultures* (1999).

³ It should be noticed that this practice-based definition of project does not necessarily coincide with one single funding source –typically one project benefits from diverse grants and/or fellowships.

⁴ Vertical 'collaborations' between principal investigators and various bench researchers (typically postdoctoral or postgraduate), with the associated divisions of labour, were indeed found in all the cases examined (Laudel, 2001, pp. 765-767). They will be addressed as 'recruitment' strategy, when involving researchers joining a laboratory with an 'external' expertise.

on previous studies (Palmer, 1999; Sanz-Menéndez et al., 2001; Laudel, 2001), we hypothesize that there are various possible forms and rationales for knowledge acquisition.

Although a research project is tightly associated with the technical and human resources available in the laboratory (which are dependent on previous achievements), the distinction between the laboratory and the project is helpful to capture knowledge integration accurately. This is because in biosciences laboratories there are often various lines of research being pursued by different intra-laboratory teams, without this necessarily leading to synthetic efforts among them.

The choice of bionanoscience as the empirical object of study responds to its current funding importance and perception as a particularly interdisciplinary field. The UK's BBSRC (2006) defines bionanotechnology as 'a multi-disciplinary area that sits at the interface between engineering and the biological and physical sciences', while the OECD defines it as an area that 'covers the interface between physics, biology, chemistry and the engineering sciences' (OECD, 2005). Within bionanoscience, we have focused on the research specialties of biomolecular motors (BM) and lab-on-a-chip (LOC), given their explicit interdisciplinary character and relevance for the NBIC vision.

The BM research specialty studies the motor proteins (myosin, kinesin, dynein, F₁-ATPase and others) that generate force at sub-cellular level using the chemical energy stored in bio-molecules (Schliwa and Woehike, 2003). BMs are essential for the movements in living beings, for example in muscle contraction, transport of vesicles in neurons or propulsion of bacterial flagella. Given their size and source of energy (small chemical molecules), BMs are viewed as potential 'engines' or actuators for nano-devices, as well as being of interest for understanding a variety of diseases. LOC research investigates construction or use of devices that integrate microfluidic and analytical functions in a small chip (Whitesides, 2006). LOC devices typically consists of a chip of solid material (glass, polymers or others depending on applications) carved or moulded

with micro/nano channels through which liquid solutions or suspensions are flowed in order to control precisely their delivery and/or to carry out chemical or genetic analyses. BM and LOC are a fundamental part of the NBIC vision as connectors between biological systems and the NBIC-engineering devices that will combine ICT, nano- and bio- technologies to enhance, substitute or repair biological functions,

For each research specialty, we conducted five case studies, each corresponding to a specific research project (as defined above). The cases were selected among the contributions by Japanese researchers in two international conferences (one for specialty) in very specific topics: mechano-chemistry of BM, and LOC devices for cells. The selection of projects from one only country working on narrowly defined topics was deliberate: it was designed to ensure that variety among the practices employed was not due to differences related to national settings or scientific or technological needs. A caveat to this investigation is the extent to which the results may be contingent on the particular specialties and period examined.

For each of the ten case studies, we conducted one interview (in some cases two) of 1-2 hours to the principal investigator and/or main contributors to the project, inquiring about the affiliations and background of the research teams, experience, main instruments, techniques and concepts used, and where they had been acquired. This data was complemented with information from scientific publications, miscellaneous documentation and homepages. A special feature our method is that the project's most relevant research articles were analysed to prepare the interviews. This allowed us to make specific and technical inquiries into 'who' had contributed to a given piece of research, and 'how', in order to complement and contrast the narratives offered by the interviewees. This strategy proved particularly valuable in order to understand the synergies and division of labour among researchers.

3. Dynamics of knowledge integration

Before looking at the research strategies adopted in each project, first we examine the characteristics of each specialty to establish how they influence the knowledge integration processes; and second, we examine the key role of instrumentation and techniques in both fields.

The current research community of BM can be traced back to at least the early 1980s, with some of the leading incumbents already active at that time. The field progressively emerged in the 1960s and 1970s as a result of seminal research on muscle physiology conducted during the 1950s. The designation of the field as ‘molecular motors’ only became widespread in the late 1990s. The high scientific impact of the specialty has led to BM research diffusing to other scientific areas, such as engineering of actuators for nano-devices (or nano-engines) or applications in medicine, developmental biology and/or neurology⁵. Since BM transform energy stored in chemical bonds into work, its study involves aspects of biophysics (such as force and energy), biochemistry (binding sites), structural biology (protein structure) and cell biology (effects on cytoskeleton functions) and the use of molecular biology techniques (Schliwa and Woehike, 2003). Hence it is viewed as needing interdisciplinary efforts.

The LOC research specialty developed over the last two decades, boosted by demands in molecular biology (e.g. high throughput DNA sequencing) and defence (biological and chemical weapons) (Whitesides, 2006). Research on LOC combines techniques from microelectronics and micro-electro-mechanical systems (MEMS) (e.g. photolithography), analytical chemistry (e.g. capillary electrophoresis), microscopy (e.g. scanning probe microscopes), and, depending on the samples being analysed, knowledge from various biology fields. There have been regular conferences held since 1994 (annually from 2000) and a specialist journal was launched in 2001.

⁵ The research community that is developing nanotechnology applications of BM since *circa* 2000 is different (although with some overlap) to the basic research community presented here. Interestingly it has some of the characteristics of LOC community: more distance between disciplines involved and knowledge transfer through service collaboration of standardised methods and materials.

Although both research specialties clearly fall into bionanoscience/technology, their dynamics, over the periods we analysed (1992-2003 for BM and 2000-2006 for LOC), were remarkably different. First, research in BM was aimed at understanding a natural phenomenon whereas LOC is by definition a technological problem, the goal being the fabrication of devices. Second, whereas the research community for BM was well established, the LOC community, even in the early 2000s, appeared socially more fragmented, with less coherent lines of research.⁶ Third, whereas the search space for new techniques or approaches in BM was within a broad area of related biosciences, which are often in contact and may share at least part of their epistemologies (e.g. biophysics, biochemistry and cell biology), LOC needed to combined knowledge from usually unrelated areas of science (e.g. electronics, cell biology and analytical chemistry), whose joint development posed major challenges (see Figure 1). A recent science map of hot research areas highlights the central and detached position of the ‘Chemical and bio-systems with microchips’ research area, between the life sciences and physico-chemical areas, filling a large gap between ‘Post-genomics’ and ‘Nanomaterials/devices’ (NISTEP, 2007, p. 55).

Table 1. Differences in research dynamics between biomolecular motors and lab-on-a-chip

	Biomolecular motors (92-03)	Lab-on-a-chip (00-06)
Type of problem	Scientific (understanding nature)	Technological (engineering devices)
Research community	Long and well established	New and forming
Combination of knowledge	From related fields (biophysics, biochemistry, cell biology)	From distant, unrelated fields (electronics, MEMS, analytical chemistry, cell biology, biochemistry)
Research challenge	Development of new technologies to tackle scientific problems	Integration of existing technologies for new applications

Finally, in both research specialties development of technology plays a central role⁷, but in different ways. In BM, new measurement and visualisation techniques (e.g. single molecule microscopy and nano-manipulation) needed to be developed in *depth* in order to

⁶ This fragmentation can be seen, for example, in the proportion of common referencing in different research efforts.

⁷ One issue deserving further exploration that is not discussed here is the apparently dominant role of technological over theoretical contributions in the development of these research specialties.

tackle a scientific problem – hence this was a case of science-driven technological development. In LOC, however, some of the most central techniques (e.g. nanofabrication and cell culture) were very well established, and the challenge was to integrate the *breadth* of techniques into a single chip. This integration process created new problems which in turn originated new technological demands (e.g. biocompatibility or cell attachment to surfaces).

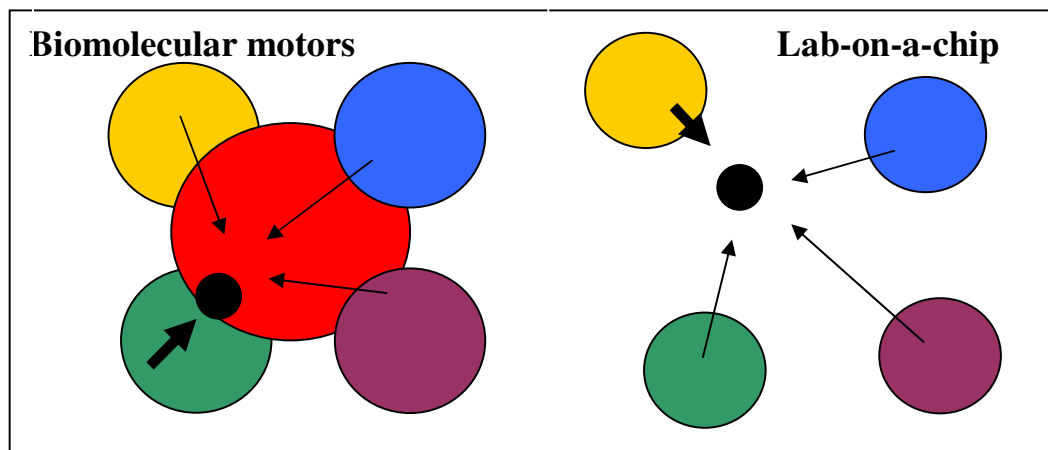


Figure 1. Differences in the knowledge search of research specialties. The black spot represents the research specialty under study. Left: In biomolecular motors (BM), a research project combines related, overlapping areas of knowledge, generally with a dominant base of expertise. Right: Lab-on-a-chip (LOC) research uses techniques from unrelated areas of science. One project also draws on one main source of expertise.

Major commonalities among the processes of knowledge integration in the two research specialties are the breadth and asymmetry in the portfolio of expertise used within one project. In all the projects examined, researchers used a wide range of techniques with distinct disciplinary bases, but at various degrees of expertise. Projects were built around one to three techniques or instruments in which the home laboratory excelled, with complementary contributions from other instruments or techniques in which they had a much lower degree of expertise. For example, one BM research laboratory made a breakthrough in nano-manipulation and single molecule microscopy while using protein preparation techniques that were very common among the BM research community. And various laboratories in LOC used of standard cell culture techniques while focusing their attention in the integration of their own microscopy technique (e.g. thermal lens

microscopy or scanning electrochemical microscopy) within the constraints of the chip. Some of these techniques, whose use requires less expertise because they have become more reliable through standard protocols, kits, and instrumentation, can be viewed as Fujimura's (1992) standardised packages or 'grey boxes', which greatly facilitate knowledge 'translation' among research specialties.

In summary, the laboratories developed and operated one to three techniques or instruments at the frontier stage making important and consistent investments of time and/or capital, but applied many other techniques at much lower level of sophistication, without making efforts to achieve technical excellence. Through this combination of specialised and standard expertise laboratories progressively created new lines on research that built on previous achievements.

4. Strategies for knowledge acquisition

Although policy discourse often assumes that interdisciplinarity research is achieved through collaborations between researchers from different backgrounds, as explained in the introduction, scholarly investigations have demonstrated that there is a rich variety of practices or research modes involved in interdisciplinary outcomes. Palmer (1999) identified four modes of interdisciplinary research (which are not mutually exclusive):

- (i) a leader recruiting a diverse team;
- (ii) a collaborator consulting colleagues;
- (iii) an individualistic researcher pursuing an integrative agenda;
- (iv) problem-oriented research combining the first three modes, in an ad-hoc manner;

Palmer suggests that each mode involves three main knowledge acquisition strategies: (i) recruiting ; (ii) consulting and (iii) learning. Using similar categories to analyse interdisciplinary fields, Sanz-Menéndez et al. (2001) found that laboratories with less diverse research expertise tended to collaborate more than laboratories with very mixed expertise, which could find the required diversity among colleagues.

Laudel (2001, 2002) showed that there are different types of collaborations, with different associated reward and communication practices:

- (i) collaboration involving division of labour;
- (ii) service collaboration;
- (iii) transmission of know-how;
- (iv) provision of access to research equipment;
- (v) trusted assessorship;
- (vi) mutual stimulation

In this study we have used this typology of collaboration to analyse the practices for knowledge acquisition employed in the projects investigated. Given that the time period of our analysis was between two to five years, in most cases studies several of the strategies identified were used during a single project. For the sake of parsimony, Palmer and Laudel's frameworks are simplified into four types of knowledge acquisition strategies: (i) service collaboration, which includes provision of research materials, but also access to equipment, assessorship and transmission of know-how; (ii) collaboration with division of labour but jointly developed research goals, which we will label as 'deep collaborations'; (iii) recruitment of researchers with complementary expertise; (iv) in-house learning efforts, which often follow some form of collaborative strategy allowing the transmission of know-how in the short term. It should be remembered that we only consider collaborations to be interactions between laboratories; knowledge exchange between researchers within an individual laboratory is seen as a result from the recruitment strategy.

4.1 Service collaboration

In service collaboration one laboratory supports the research of another one by providing materials, access to instrumentation, assessment regarding data interpretation, the use of materials or instruments, or general know-how. Although these activities can be considered separately, as in Laudel (2001), in the cases we studied, they were often

intertwined (e.g. a medical doctor providing cells for a LOC experiment explained how to culture them and helped check their viability in the initial stages of the research). The providers of the service side of the collaboration may or may not be co-authors of the eventual paper depending on their personal relations with the researcher leading the project or the field's norms.⁸ But independently of whether they are granted the co-authorship or not, the researchers providing the services act only on a supportive or advisory capacity and do not participate actively in either the design or the final interpretation of the results,

Service collaboration seems to have been extremely important for getting the LOC studies started. This is because, in the initial stages, researchers had little or no expertise in some of the materials and instruments they wanted to 'try out'. The service part of the collaboration generally did not spend time trying to understand the details of the research project, but could quite easily help design, demonstrate how to follow routine experimental procedures and give advice on the application of techniques.

In one example of service collaboration in LOC, Professor X, in a the graduate school of environmental sciences, with a background in electrochemistry and expertise in scanning electrochemical microscopy (SECM) wanted to create a microchip to monitor cell reactions to specific chemical environments (possible application in pollution or pharmaceutical testing). Since all the students and researchers in his laboratory had backgrounds in applied chemistry, he needed two external sources of expertise: one in fabrication of the microchip, and the other in cell culture. Over the years, Professor X's laboratory had progressively acquired the expertise (and later also bought the equipment) required for micro-fabrication from a large MEMS laboratory in the same university, which had the expensive equipment and was willing to provide expert guidance on its use via its PhD students to 'outside' researchers. In addition, Professor X's PhD students acquired knowledge and expertise in cell culture and simple molecular biology techniques, from a medical doctor who was a member of a cross-departmental

⁸ Regarding co-authoring, we have found differences even in projects with very similar topics, where one would have expected the same social norms to apply.

programme. Hence, this is a case, and is probably a quite common occurrence, of one rather focused and homogeneous laboratory building on its expertise (SECM) and acquiring materials, instrumentation and know-how through a service collaboration with colleagues within the same academic institution. Although the contributions of the collaborators were essential, their investment of time and resources was limited – as shown by the fact that they were not co-authors of or even acknowledged in the eventual publications of the research.

In BM, however, service collaborations do not appear to have played a major role. The only important example is the case of a mixed cell biology-biophysics laboratory that provided a molecular engineered protein to a biophysics-based group which had developed a new type of fluorescent microscope. Other instances of service collaborations in BM (e.g. as provision of standard electron-microscopy or consultations regarding nano-manipulation) were rather complementary, less relevant contributions to the projects. It could be argued that in BM, the research community already shared a body of knowledge, and the types of expertise that might lead to service collaboration, had either been absorbed internally in the laboratories or could be more easily transmitted through informal contacts. Also, BM research did not generally require the use of large and expensive equipment.⁹

4.2 Deep collaboration

Deep collaboration is defined here as a research project that is jointly formulated and interpreted by the collaborating laboratories, but where each of the groups involved makes an essential contribution to different stages of the research process; hence it involves division of labour yet the collaborators interact strongly (i.e. ‘deeply’). This is the type of collaboration associated with interdisciplinary research in ‘conventional wisdom’. It was found to be present in all the cases of BM research and to contribute to a broadening of the cognitive base (as would be expected) by increasing the diversity of

⁹ This has changed in some areas, in particular in those using micro- or nanofabrication techniques to conduct BM experiments.

researchers' backgrounds and the available expertise in instrumentation. However, in only one case was it the most important strategy. In most cases, first, the 'home' laboratory made a new major contribution to the field, and second, a deep collaboration extended this contribution by adding techniques and expertise from another team.

Another feature of the deep collaborations found in BM is that they generally occurred among laboratories belonging to the BM research community, but based on different disciplines (e.g. biophysics and structural biology). This pattern suggests that the previous sharing of interests, scientific language and social networks that occur within research specialties, are important to facilitate the mutual understanding necessary for deep collaboration. Having said this, the only case study where deep collaboration was observed across research specialties (linking bioenergetics and research in linear BM), is precisely the case in which this is the main knowledge acquisition strategy and the project had the most impact (as shown by citations). This (but only this) instance supports the widespread idea that deep collaborations bridging longer cognitive distances may be much riskier (higher chances of failure) but may also result in path-breaking contributions (higher impact if successful).

To our surprise, we did not find any examples of deep collaboration in LOC research in the period studied (early 2000s), but this is changing as this field becomes more mature. A Professor of electronics explained the evolution: when he first moved into bionanotechnology in the late 1990s he would have been happy to collaborate with biologists, but did not know how to engage them, and at that time did not have the prestige required to catch their attention. Only once he had succeeded in creating some LOC prototypes (and was in receipt of a large grant), biology-based researchers became interested in collaborating with him for specific applications. This example illustrates a pattern we have observed in various case studies: deep collaborations extend to collaborators the use of techniques that have previously proved valuable.

4.3 Recruitment of researchers with distinct backgrounds

Recruitment of researchers with diverse backgrounds is a way of introducing ‘external’ knowledge into a laboratory. In BM, we found this to be an important strategy in three of the five cases investigated. As in most of the cases of deep collaboration, though, interdisciplinary recruitment of researchers was based on both complementarity and overlapping of expertise regarding techniques and/or research problems: (i) a biochemist, with experience in BM, joined a mainly biophysics laboratory in BM to contribute his protein purification expertise; (ii) a laboratory had a recruitment strategy balancing the number of biophysicists and cell biologists; (iii) one laboratory consisted mainly of biophysicists, but some of them had a somewhat diverse technical expertise.

In LOC research, again a different picture emerged. One of the case studies was located in a new centre for biomedical applications (opened in 2005) with a large laboratory shared by researchers with different backgrounds, engaged in different lines of research. Most of these researchers were based in the centre, but some also had their own laboratories in another department within the same university. The case study examined involved a young researcher (biochemist by training), who had previously worked on the development of molecular biology techniques, and was now developing a LOC for cell analysis working together with an analytical chemist with expertise in electrophoresis and a neuro-scientist interested in neurite navigation. Their combined expertise was allowing them to develop most of the research on their own, with the exception of the microchip, which was provided through service collaboration by another colleague from the centre who had micro-fabrication equipment in his own laboratory. This case fits with the vision of nanobiotechnology as being developed in transient and fully interdisciplinary teams. Interestingly, it was the most recent of all the cases investigated and was based in a new organisation trying to create a ‘new’ institutional culture, which also had other Mode-2 features (e.g. an emphasis on patenting and secrecy).

In the other four cases of LOC, however, there was much less mix, and more similarity with the findings for BM. The teams of the various projects included:

1. A 'pure' biology PhD student in a biophysics laboratory.
2. A Professor in analytical chemistry hiring a postdoctoral researcher with a background in biochemistry at a food science department.
3. A team was based around electronics, with all the key contributors belonging to this discipline, but with some among them with unique expertise: one contributed the setup of a sensor, another the polymer coating, and others contributed to the experimental setup.
4. An electro-chemistry laboratory that did not recruit researchers from different backgrounds.

The importance of having both overlap and complementarity in recruitment was highlighted by a Professor of electronics in one of the LOC cases. In his starting years in LOC, he had not hired researchers from biology fields because his lack of understanding of these fields meant that he had no criteria for selecting candidates, and also there were no young researchers with adequate hybrid backgrounds with whom he could relate to. But the situation had recently changed, he argued: both his capability to judge biologists in LOC and the availability of young researchers with some experience both engineering and biological aspects of LOC, had increased in the very last years.

The downside of recruitment as strategy is that it is highly dependent on funding. Two of the LOC laboratories had only recruited researchers with complementary expertise (a cell biologist in an electronics-based laboratory, a pharmacologist in a biophysics-based laboratory) after receiving large grants following initial successes in the field. A third one wanted to, but didn't have the funding.

4.4 In-house learning

The search, identification, acquisition and integration of 'external' knowledge into a laboratory involve important learning efforts. These processes have been conceptualised as 'interpretative learning' (Grigg et al., 2003), or as the acquisition of *absorptive*

capabilities in the innovation literature, allowing laboratories ‘to recognize the value of new, external information, assimilate it, and apply it’ to specific goals (Cohen and Levinthal, 1990, p. 125). The importance of these in-house learning processes should not be underestimated.

In service collaborations, when researchers use materials or instruments obtained from collaborators, their degree of conceptual or theoretical engagement with those materials may be limited, but they often have to acquire the minimum knowledge to set up an experimental base for their application. For example, two electronics laboratories in LOC used cells provided to them through service collaboration, but in order to do so, they had to learn how to make their own cell cultures. Thus, they developed in-house standard expertise, and now each of these laboratories has a room that houses incubators, microscopes, autoclaves, shakers and centrifuges, and at least one researcher who is familiar with the protocols for preparing cells.

In the case of deep collaboration as well, the participating laboratories need to be familiar with each other’s contributions and procedures – which can be difficult when tacit knowledge is involved. In one of the cases of deep collaboration in which one laboratory prepared an engineered protein motor and the other carried out the X-crystallography, after repeated failures, a researcher from Tokyo had to be flown to San Francisco to teach *in situ* how to prepare the protein sample. In the case of recruitment, there are as well learning processes resulting from the interactions among recently arrived researchers and the researchers embodying the domestic expertise of the hosting laboratory. Finally, researchers can also learn from codified knowledge sources, such as publications and conference presentations.

The degree of in-house learning seems to depend on laboratory goals and institutional factors. Those laboratories with an educational focus that are well established and funded, and/or can work towards long-term objectives, are in a better position to engage in development and internalisation of new expertise. A BM project in the medical school of a very prestigious university illustrates this point. This laboratory was well-funded

laboratory and relied almost exclusively on students with a medical background yet it dealt with a surprisingly wide diversity of topics and disciplines. This cognitive diversity could be achieved because the leading professor was willing to provide time and support to students to learn state-of-the-art techniques in fields in the neighbourhood of his 'domestic' cell biology base. Thus, in the project we analysed in this centre, one PhD student collaborated briefly with an expert on cryo-electron-microscopy and then went on to conduct his own research using the expertise he had acquired, and another postdoctoral researcher spent a long-time developing fluorescent microscopy (at single molecule resolution) with the help of informal consultations with a major microscope maker. The professor's view was that although the learning processes were lengthy, the incorporation of these techniques into the laboratory would pay off in the long term, both for the laboratory and for the young researchers involved.

Table 2. Summary of strategies for knowledge acquisition observed, and conditions facilitating their occurrence (as inferred from case studies).

Strategies	Definition	OBSERVATIONS FROM CASE STUDIES		Conditions that facilitate occurrence
		Biomolecular motors (BM) (five cases)	Lab-on-a-chip (LOC) (five cases)	
Service Collaboration	Provision of materials, access to instrumentation or transmission of know-how.	NOT significant in four cases. Relevant in one case (one lab sends engineered protein to microscopy-focused lab).	Very important in ALL cases. Laboratories borrow cells lines, nanofabrication facilities, etc. and need associated know-how.	Large cognitive distance between labs. Existence of standardised packages of methods and materials. Lack or costly access to materials or instrumentation.
Deep Collaboration	Projects jointly formulated and interpreted by the collaborating labs, but typically carried out with division of labour.	One case where it is essential for breakthrough. In three other cases, it is relevant, but it follows major achievements developed in-house.	NOT present in this period. It begins to occur ONLY after the labs achieve some expertise beyond their domestic knowledge base.	Short to middle cognitive distance between collaborating labs in order to keep coordination costs low. Labs belong to the same research community/research problem, but with a different disciplinary focus.
Recruitment	Employment of researchers with backgrounds that differ from the lab's knowledge base.	One case of mixed lab but among related disciplines. Important in three cases , but to a limited degree. 'External' recruits not only complement but also overlap the knowledge base of the hiring lab.	One case with recruits from many disciplines. In three other cases, also present to a limited degree (just one 'external' recruit or complementary skills within same discipline). It becomes more frequent in later period.	Short to middle cognitive distance between hiring lab and recruit. Access to funding. Recruit has both overlap and complementary with the hiring lab.
In-house Learning	Development of practices previously external to the lab's domestic knowledge base	Essential in two cases , in which the labs are based mainly in one body of knowledge. Relevant in all cases , as a requisite for collaboration.	Relevant in all cases , given that service collaboration requires some understanding of 'external' materials and methods.	In-house learning is needed so as to be able to adopt the materials, methods received from collaborators. (Absorptive capacity)

5. Discussion and conclusions

5.1. Summary of empirical findings: various strategies for knowledge acquisition

In this paper we have investigated interdisciplinary practices for ten case studies in two different specialties in bionanoscience (biomolecular motors (BM) and lab-on-a-chip (LOC)), through analysis of the dynamics of knowledge integration and the strategies of knowledge acquisition carried out in a given research project by one laboratory.

First, we found that knowledge integration in a research project was asymmetrical. In both research specialties, projects were carried out with contributions from different bodies of knowledge, and hence the laboratories involved needed to use a breadth of techniques from various research traditions. However, the depth of expertise was very unequal: a project typically focused on the unique know-how of its home-laboratory in a particular research method or instrument, but used standard expertise in many other techniques. This asymmetry was more prominent in LOC, where the bodies of knowledge being integrated were more distinct. Some of these sophisticated but well-established techniques transferred between laboratories may be viewed as ‘standardised packages’ (Fujimura, 1992).

Second, we found that various strategies were used to acquire expertise from different bodies of knowledge (see Table 2). Although deep collaboration (among laboratories from different disciplines) or mixed laboratories (with researchers from many disciplinary backgrounds) are presumed to be the essential strategies in convergent technologies, in only two projects of the ten examined were these strategies dominant. In the other eight case studies, other strategies were more important: (i) service collaboration (particularly in LOC), (ii) recruitment of a limited number of researchers with complementary expertise (but not necessarily from a different discipline), (iii) and in-house learning. Projects generally relied on more than one strategy for knowledge acquisition.

In BM, a long established research specialty, recruitment and collaborations were common and occurred mainly between researchers and laboratories that were part of the same or related research communities but had different core expertise (e.g. nano-manipulation vs. electron microscopy). In LOC, however, three of the five cases studies did not show either deep collaboration or recruitment in the period examined –but two out of these three, began to use these strategies as they accumulated ‘hybrid’ knowledge and the field itself became more mature. These observations suggest that, although both recruitment and deep collaboration are adopted with the objective of acquiring complementary expertise, they are difficult to establish without the existence of an overlapping body of knowledge. Service collaboration, which does not require this common knowledge base, was crucial in the early stages of LOC research.

Finally, we found that knowledge practices are highly dependent on the goals and institutional settings of the laboratory such as type of funding, presence of postgraduate students, etc., which make it possible or not to invest time, human, technical and social resources in in-house learning, recruitment or collaboration. For example, we found that a mid-term project with high funding relied on recruitment, whilst some laboratories with many postgraduates tended to internalise new knowledge through short time collaborations followed by in-house development. The effect of institutional constraints on laboratory practices is an aspect that was only tangentially touched upon in this study and deserves further exploration.

5.2. A trade-off between cognitive diversity and cohesion

At the light of these observations, one may ask: Why are the knowledge acquisition strategies not as ‘fully’ interdisciplinary as might be expected from the policy discourse? And under what conditions are some knowledge acquisition strategies more appropriate than others?

To answer these questions, we follow Katz and Martin's (1997, p. 17) suggestion of adopting a 'symmetrical approach' in assessing both the benefits and the costs of interdisciplinarity. Hollingsworth (2006) provided some evidence that major biomedical discoveries have occurred in organisations that combined moderately high cognitive diversity with high social cohesion. However, the literature on scientific and inter-firm collaboration has argued that there is a tension between cognitive diversity and coordination, i.e. between 'the benefits to innovation of working across disciplinary and organizational boundaries versus the risk that arise from the costs of coordination and relationship development in these collaborations' (Cummings and Kiesler, 2005, p. 704). Similarly, Llerena and Meyer-Krahmer (2004) presented this argument in terms of a trade-off between the benefits of cognitive diversity, and the learning and cohesion costs. As a result of this trade-off, the performance of projects is thought to be higher at moderate levels of diversity.

This cohesion-diversity trade-off model may explain some of the practices observed, in particular, why high and distant cognitive mixes are rare.. In BM, collaboration and recruitment were facilitated by an overlap in the knowledge base between laboratories and researchers, which kept coordination costs low while still providing some gains in cognitive diversity. In LOC, however, deep collaboration and recruitment had to bridge major cognitive distances; in such conditions the costs of cohesion were greater than the cognitive gains from diversity –and thus these strategies were not adopted. We observed, instead, that the LOC laboratories tended to engage in service collaboration, which provided limited knowledge diversification, but required little organisational effort. Moreover the possibilities of establishing service collaborations were enhanced by standardised materials or methods which facilitated their adoption or translation in new research communities (Fujimura, 1992).

A richer understanding of the choice of knowledge acquisition strategy needs to consider as well goals and institutional settings of projects: how and why laboratories with many postgraduate may favour in-house learning as part of their education, or how service collaborations are facilitated by funding support from governmental initiatives (e.g. in

several countries ‘open-door’ nanofabrication facilities are publicly funded so that ‘external’ researchers can use them).

5.3 Institutional frameworks for interdisciplinary research

As described in the introduction, discourses on new modes of knowledge production claim that fields such as bionanotechnology require interdisciplinarity (meaning knowledge integration), which in turn, requires certain forms of organisation (e.g. deep collaborations). We found out that bionanoscience projects involved knowledge integration, but that laboratories actually engaged in diverse knowledge acquisition strategies that most often do not maximize the cognitive diversity or the number of disciplinary affiliations or backgrounds – an observation that is understandable if coordination costs are taken into account. After these findings, is there a case for interdisciplinary laboratories and collaborations still be fostered in bionanoscience?

This study has been a *descriptive* analysis of how research in bionanotechnology was carried out in the late 1990s and early 2000s in two research specialties. The fact that during these periods, laboratories engaged little in deep collaboration and recruitment across research fields does not mean that under more favourable institutional frameworks they could. In other words, since our research was not prescriptive, we neither proved nor disproved that institutional changes fostering interdisciplinary could have positive impacts in emerging fields. What our findings highlight is that there are various organisational practices that can support integrative efforts. This observation challenges the view that the scientific needs of bionanoscience necessarily require specific organisational structures. In doing so, our results suggest that support for integrative (or convergent) sciences and technologies can be delivered by a variety of policy instruments and support the analysts who critique the discourses on interdisciplinarity and ‘new modes of knowledge production’ as performative (Godin, 1998), or normative (Weingart, 2000).

Hence, our findings raise the question of whether some of the current policy instruments, focusing on formal collaborations between laboratories, are the most appropriate to facilitate knowledge integration. Complementary policies, such as instrumentation platforms or small grants for short term technical exchanges that facilitate service collaboration, may play as well a positive role.

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